ABSTRACT OF MAJOR PROJECT

DATE:

BRANCH:CAI-A

TEAM NUMBER:16

FinRAG AI — Integrated Financial Intelligence Platform is an AI-powered financial intelligence system that combines predictive transaction analytics with metadata-driven Retrieval-Augmented Generation (RAG) for financial document question answering. Financial institutions handle both large volumes of structured transaction data and complex unstructured documents such as annual reports, financial filings, and corporate disclosures. However, these data sources are often analyzed independently, making financial investigation time-consuming and limiting the availability of contextual evidence.

The proposed system integrates two complementary intelligence engines within a unified platform. The Transaction Intelligence Engine uses machine learning and anomaly detection techniques to analyze transaction behavior, identify suspicious patterns, calculate risk scores, and generate alerts. The Financial RAG Engine processes financial documents using document parsing, LLM-generated metadata, contextual embeddings, dense and BM25-based hybrid retrieval, and reranking to retrieve relevant financial evidence. A large language model then generates grounded answers based on the retrieved context.

The platform provides a unified dashboard through which users can investigate transaction risk and simultaneously access supporting financial-document evidence. The RAG component is based on recent research on metadata-driven financial question answering and is evaluated using metrics such as Claim Recall, Context Precision, F1-score, Faithfulness, and Hallucination. The transaction intelligence component is evaluated using Precision, Recall, F1-score, ROC-AUC/PR-AUC, false-positive rate, and detection latency.

The proposed system aims to demonstrate how Machine Learning, Anomaly Detection, Natural Language Processing, RAG, Large Language Models, Information Retrieval, Vector Databases, and Full-Stack Development can be integrated into a practical financial intelligence platform for faster, more explainable, and evidence-grounded financial analysis.